



ZIMBABWE

**MINISTRY OF HIGHER AND TERTIARY EDUCATION,
INNOVATION, SCIENCE AND TECHNOLOGY
DEVELOPMENT**

**HIGHER EDUCATION EXAMINATIONS COUNCIL
(HEXCO)**

NATIONAL CERTIFICATE

IN

DIESEL PLANT FITTING

MODULE: Engine and Plant Maintenance

PAPER NO: 319/22/M02

DURATION: 3 Hours

NOVEMBER/ DECEMBER 2025 EXAMINATION

REQUIREMENTS

Pen, Drawing Instruments and Answer Booklet

INSTRUCTIONS TO CANDIDATE

- 1. Section A: Answer all questions (20 marks).**
- 2. Section B: Answer all questions (20 marks).**
- 3. Section C: Answer all questions (60 marks).**

This paper consists of 5 printed pages.

SECTION A

QUESTION 1

- a) State six (6) causes of Engine overheating. (6 marks)
- b) Give the name of the thermostat which fails safe. (2 marks)
- c) Engine Cylinder Compression is tested using a (2 marks)
- d) State two (2) functions of the waste gate valve on the turbocharger. (4 marks)
- e) Give four advantages of a direct Injection type of Combustion Chamber. (4 marks)
- f) Define the following terms:
 - i) Volumetric Efficiency (1 mark)
 - ii) Mechanical Efficiency (1 mark)

SECTION B

QUESTION 2

- a) Give the name of the piston ring which is used as a top Compression ring in a worn-out Cylinder bore. (2 marks)
- b) A faulty injector on an engine would cause (1 mark)
- c) Define dieseling or after running. (2 marks)
- d) State three (3) functions of lubricating oil. (3 marks)
- e) State two (2) firing orders of a six Cylinder engine. (2 marks)
- f) Define the following engine terms.
 - i) Ineffective crank angle (1 mark)
 - ii) Crank angularity (1 mark)
- g) List three (3) possible causes of black smoke from the engine (3 marks)
- h) List two positions where we can find camshaft on an engine. (2 marks)
- i) State three methods of achieving tappet rotation. (3 marks)

SECTION C

QUESTION 3

- a) Draw a well labelled valve and port timing diagram of the Uniflow two stroke Detroit Diesel engine. (8 marks)
- b) State the position where valve and port overlap takes place. (2 marks)

QUESTION 4

Draw a well labelled diagram of a poppet valve. (10 marks)

QUESTION 5

- a) Give the name of D.P.A pump in full. (2 marks)
- b) State the name of the component which act as an anti-dribble device on the D.P.A pump. (2 marks)
- c) List two types of governors used with this type of pump. (2 marks)
- d) State two functions of the pressure regulating valve. (2 marks)
- e) Give two types of metering valves used in connection with this pump. (2 marks)

QUESTION 6

- a) Describe how each of the following is formed in the engine lubrication system and state two effects of each.
- i) Oil dilution (4 marks)
 - ii) Oil oxidation (4 marks)
- b) Describe how sludge is formed in the engine. (2 marks)

QUESTION 7

- a) State three materials which are used in the manufacturing of shell bearings. (3 marks)
- b) List two methods which are used to control engine temperature. (2 marks)
- c) State two advantages of using a viscous fan in the cooling system. (2 marks)
- d) List 3 advantages of a liquid cooling system. (3 marks)

QUESTION 8

State the two (2) functions of each of the following engine components.

- a) Crankshaft (2 marks)
- b) Camshaft (2 marks)
- c) Valves (2 marks)
- d) Piston (2 marks)
- e) Cylinder block (2 marks)

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